Arun Mallikarjuna

FULL-STACK ENGINEER · APPLIED AI & AUTOMATION

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PROFILE

Full-stack engineer with 5 years building production systems for telecom operators: React and Next.js frontends, Node.js APIs, dashboards, reporting and databases. I have also spent 2+ years building LLM-powered tools and automation with the Claude and Gemini APIs. I turn business problems into self-serve tools, and I publish my own developer tools, including a reporting engine on npm and a multi-agent AI code editor.

EXPERIENCE

Tecnotree Convergence · Bengaluru Oct 2021 – Present

Senior Software Engineer Jan 2024 – Present

- Built Node.js REST APIs, reporting dashboards and third-party integrations for the promotion platform, with delivery tied to explicit business and customer outcomes.
- Helped keep the platform at 99.5% availability through incident response and production debugging across distributed microservices.

DPM, Promotion Management Platform BUSINESS SELF-SERVICE TOOL

Next.js 16 (App Router) · React 19 · TypeScript · Node.js · Tailwind CSS · Zod · NextAuth

- Built the criteria and category modules, a nested AND/OR rule builder that lets business teams launch discounts, bundles and campaigns themselves, with no code change or redeployment.
- Automated bulk import, export, date-extension and lifecycle jobs, with step-by-step flows and per-transaction status, so promotions and coupons can be updated at scale.
- Built a cart simulator, calendar scheduling and an audit-log version comparator so teams can test changes before go-live. Cut frontend load time by about 30%.
- Implemented SSO across WSO2 and ForgeRock OIDC with role-based access. Automated delivery with Jenkins CI/CD, SonarQube and Trivy gates, and Rancher deploys.

Software Engineer Oct 2021 – Dec 2023

- Built high-throughput Node.js REST APIs for eligibility evaluation, using async concurrency to cut latency per request. Turned customer problems into configurable workflows that non-technical teams run themselves.

SELECTED PROJECTS

ReportWright: open-source reporting engine, published on npm github.com/MrArun005/reportwright

TypeScript · Node.js · Web Workers · HarfBuzz (WASM) · React / Vue / Svelte / Angular

- Built a drag-and-drop report designer, an embeddable viewer and an engine that exports to PDF, Excel, Word, PowerPoint, HTML and CSV without Chromium. Grouped tables, pivots, multilingual text shaping, and importers for JasperReports, SSRS and BIRT. Published as 11 npm packages and covered by 374 test files.
- Extracted **@reportwright/pdf**, a streaming PDF library with zero dependencies. Memory stays flat (20,000 pages use about the heap of 1,000), output passes veraPDF checks for PDF/A and PDF/UA, and it supports PAdES signing.

HRMS: multi-tenant enterprise HR & payroll SaaS private repo · demo on request

NestJS 11 · Next.js 15 · TypeScript · MySQL (Prisma) · Redis · BullMQ · OpenSearch · Cerbos · AWS ECS · Terraform

- Built a monorepo (API, worker, web app, control plane) where each tenant has its own database, with 75+ modules spanning core HR, leave, recruitment, performance and analytics. Payroll for India and the US reads statutory rates from the database and generates NEFT/NACHA bank files and tax forms.
- Secured it with Cerbos policy-based authorization (71 policies), SAML/OIDC SSO, SCIM, maker-checker workflows and GDPR erasure, covered by 636 test files.

FriedLiver: multi-agent AI code editor github.com/MrArun005/FriedLiver

Electron · Monaco · Agent Client Protocol · VS Code (Code - OSS) fork · TypeScript

- Runs Claude, Codex and Gemini coding agents side by side. Every edit shows as a reviewable diff, every command waits for approval, and agents can hand off to or query each other. Idles at 314 MB against 761 MB for VS Code, measured with my own benchmark harness.

SKILLS & EDUCATION

LANGUAGES	TypeScript, JavaScript (ES6+), Python, SQL
AI & AUTOMATION	Claude & Gemini APIs, AI agents, prompt engineering, evals, RAG, RL (MuJoCo, PPO)
FRONTEND	React, Next.js, React Native, Tailwind CSS, Electron; dashboards & reporting UIs
BACKEND & DATA	Node.js, NestJS, Express, REST APIs; PostgreSQL, MySQL, MongoDB, Redis, OpenSearch
CLOUD & DEVOPS	AWS (ECS, RDS), GCP, Docker, Terraform, Jenkins CI/CD, GitHub Actions, Rancher, Vercel
EDUCATION	B.E. Computer Science · Maharaja Institute of Technology, Mysuru · 2021